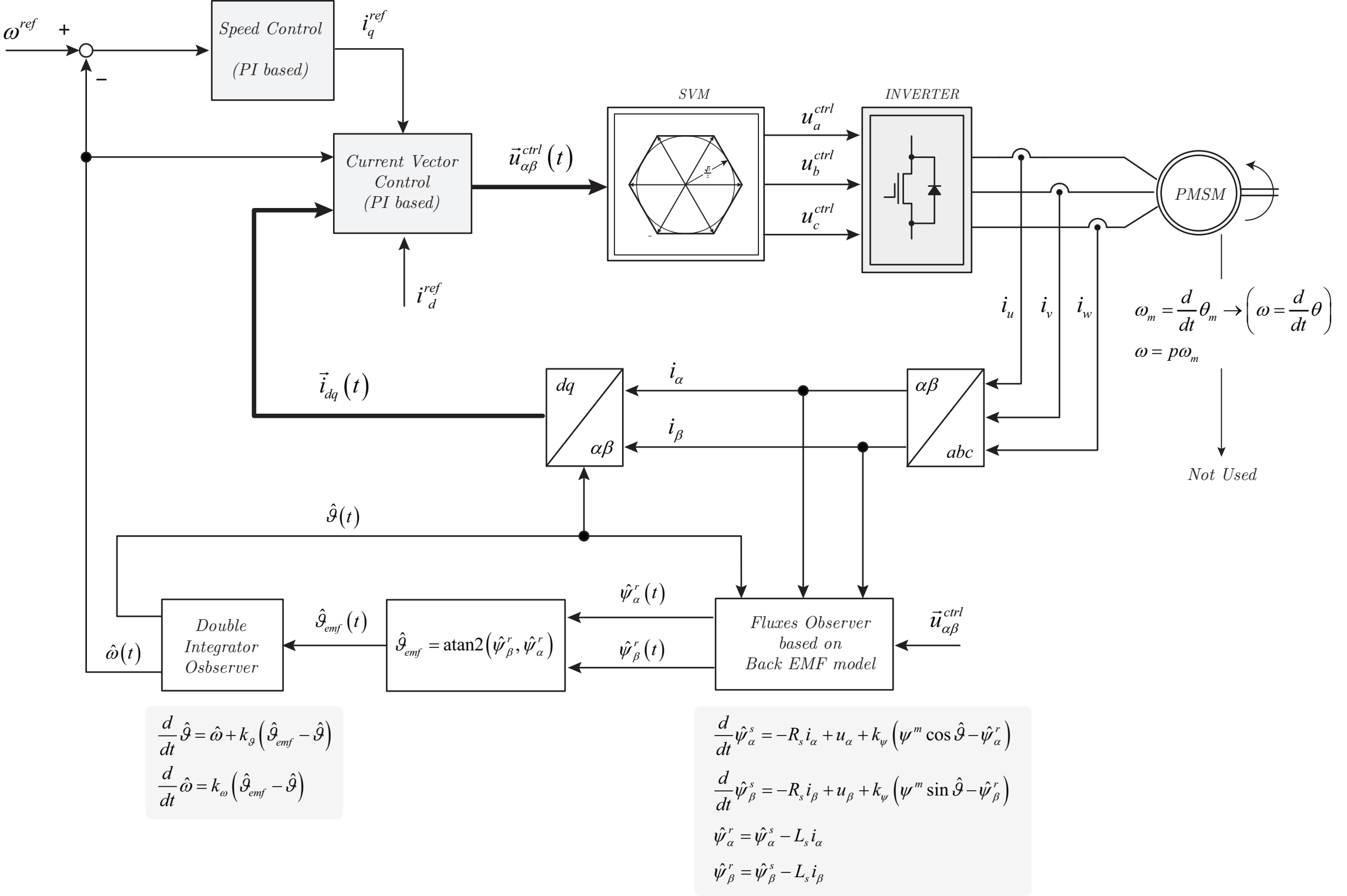
ω^{ref}
+
−
Speed Control
(PI based)
i_q^{ref}
Current Vector
Control
(PI based)
i_d^{ref}
$\vec{u}_{\alpha\beta}^{ctrl}(t)$
SVM
u_a^{ctrl}
u_b^{ctrl}
u_c^{ctrl}
INVERTER
PMSM
i_u
i_v
i_w
$\omega_m = \frac{d}{dt}\theta_m \rightarrow \left(\omega = \frac{d}{dt}\theta\right)$
$\omega = p\omega_m$
Not Used
$\vec{i}_{dq}(t)$
dq
αβ
i_α
i_β
αβ
abc
$\hat{\vartheta}(t)$
$\hat{\omega}(t)$
Double
Integrator
Osbserver
$\hat{\vartheta}_{emf}(t)$
$\hat{\vartheta}_{emf} = \text{atan2}\left(\hat{\psi}_\beta^r, \hat{\psi}_\alpha^r\right)$
$\hat{\psi}_\alpha^r(t)$
$\hat{\psi}_\beta^r(t)$
Fluxes Observer
based on
Back EMF model
$\vec{u}_{\alpha\beta}^{ctrl}$
$\frac{d}{dt}\hat{\vartheta} = \hat{\omega} + k_\vartheta\left(\hat{\vartheta}_{emf} - \hat{\vartheta}\right)$
$\frac{d}{dt}\hat{\omega} = k_\omega\left(\hat{\vartheta}_{emf} - \hat{\vartheta}\right)$
$\frac{d}{dt}\hat{\psi}_\alpha^s = -R_s i_\alpha + u_\alpha + k_\psi\left(\psi^m \cos\hat{\vartheta} - \hat{\psi}_\alpha^r\right)$
$\frac{d}{dt}\hat{\psi}_\beta^s = -R_s i_\beta + u_\beta + k_\psi\left(\psi^m \sin\hat{\vartheta} - \hat{\psi}_\beta^r\right)$
$\hat{\psi}_\alpha^r = \hat{\psi}_\alpha^s - L_s i_\alpha$
$\hat{\psi}_\beta^r = \hat{\psi}_\beta^s - L_s i_\beta$